

VOLUME 3

Mesa Conversion and Installation

Mazak VQC-20/40 SN 060231 - LinuxCNC/Mesa Retrofit

REVISION B - CONSOLIDATED FROM LIVE AUTHORITY

Field	Value
Document	MZK-LCNC-003
Revision	B
Generated	2026-08-07
Source commit	a3730f9
Machine	Mazak VQC-20/40, serial 060231
Original control	Mazatrol M-2 / MELDAS
Target control	LinuxCNC 2.9.10, Debian 13 PREEMPT_RT

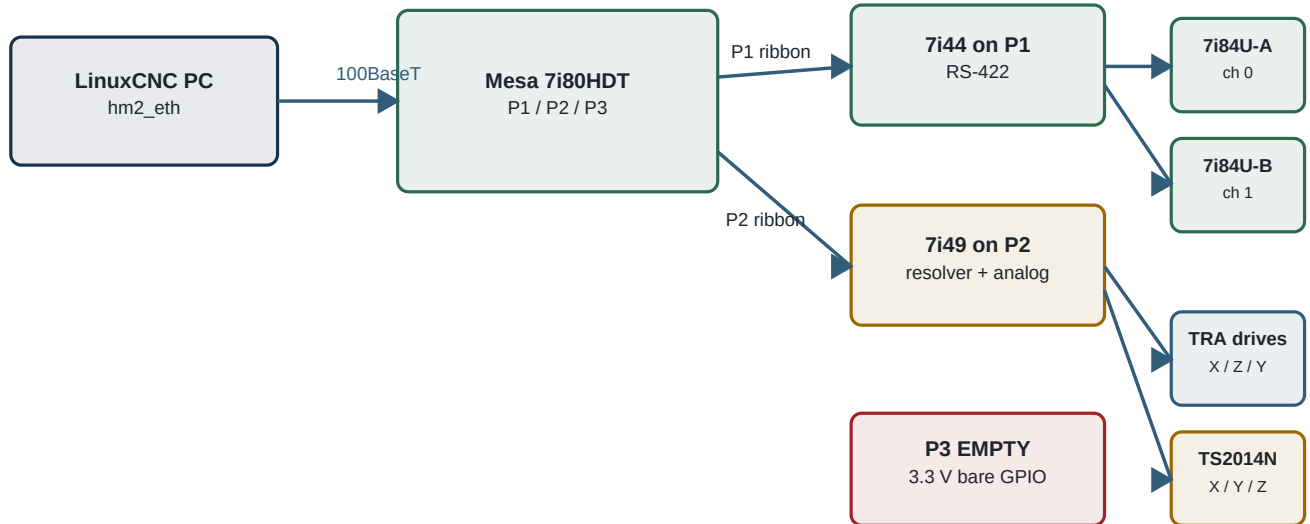
STATUS - NOT READY FOR POWER OR MOTION

The repository is still at planning/bring-up stage. No live-power hold point is signed. Authority rows marked PROPOSED, COMMISSIONING_PENDING, HOLD_CONFLICT, or UNBOUND are not as-built evidence. The hardwired E-stop chain, resolver suffixes/scales, bitfile, analog polarity/scaling, field normal states, shared-bus behavior, and spindle terminals remain to be proven.

This volume replaces Rev A. Rev A must not be used for wiring, commissioning, or service.

1. Selected architecture

The selected stack is a LinuxCNC PC connected by dedicated Ethernet to a Mesa 7i80HDT. P1 carries a 7i44 smart-serial interface to two 7i84U field-I/O cards. P2 carries a plain 7i49 for three resolver channels and four active analog outputs. P3 is intentionally empty.



Only the 7i49 carries motion feedback and analog commands. Field I/O is isolated through two 7i84Us. P3 has no field wiring.

Figure 3-1. Rev B control architecture. No 7i97T or 7i37TA is part of the selected stack.

Item	Role	Current qualification
LinuxCNC host	Trajectory, HAL, GUI	Debian 13 / LinuxCNC 2.9.10 / PREEMPT_RT target
Mesa 7i80HDT	100BaseT FPGA host, 3 x 50-pin connectors	Board/revision and loaded IDROM still to capture
Mesa 7i44 on P1	8-channel RS-422; channels 0/1 active	Physical link and exact HAL names unverified
Mesa 7i49 on P2	6 resolver + 6 bipolar analog channels	Plain board baseline; installed revision and signal levels unverified
7i84U-A ch 0	32 DI / 16 DO machine field I/O	Fully allocated; physical states unverified
7i84U-B ch 1	Limits, homes, probe, enables, relay loads	11 DI / 9 DO used; physical states unverified
P3	Future daughter-card position	Empty; bare 3.3 V GPIO must not see 24 V

2. Host and Ethernet setup

- Use a dedicated wired NIC directly connected to the 7i80HDT; no wireless or USB Ethernet.
- Board address is planned as 192.168.1.121; host control NIC is planned as 192.168.1.1/24 with no gateway. Verify the actual interface and MAC.
- Record NIC chipset, driver, firmware, negotiated 100 Mbit/s full duplex, IRQ, offload/coalescing state, and LinuxCNC startup diagnostics.
- Treat hardware-irq-coalesce-rx-usecs 0 as an A/B experiment; remove it if harmful on the installed NIC, especially affected Marvell hardware.
- Run at least 15 minutes idle latency, 4 hours worst-load latency, 2 hours with the real config and drives disabled, then cable-yank fault injection.
- Any packet-error event is investigated. packet-error-exceeded and watchdog.has_bit must latch motion inhibit and require manual recovery.

3. Firmware and IDROM gate

The working name 7i80hdt_7i44_ss_7i49d.bit is a placeholder only. Resolver configurations for the Efinix-based 7i80HDT are not proven by a stock-zip filename in this repository. Do not flash until Mesa/PCW provenance, binary, SHA-256, Efinity/source reference, recovery procedure, readhmid output, and HAL pin dump are committed as D3.

BITFILE HOLD

Never substitute a 7i80HD or 7i80HD-16 bitfile. The 7i80HDT uses a different FPGA and requires an HDT-specific build.

Expected loaded function	Planned config	Proof required
Resolvers	num_resolvers=3	IDROM + resolver.00-02 HAL pins
Analog PWM generators	num_pwmgens=4	pwmgen.00-.03 and measured AOUT mapping
Encoders	num_encoders=0	No spindle receiver/pins assigned
Smart serial	sserial_port_0=00xxxxxx	One port, channels 0/1, both remote serials
P3	IOPort only / unused	No field conductors; pin listing retained for future use

4. Resolver and analog motion path

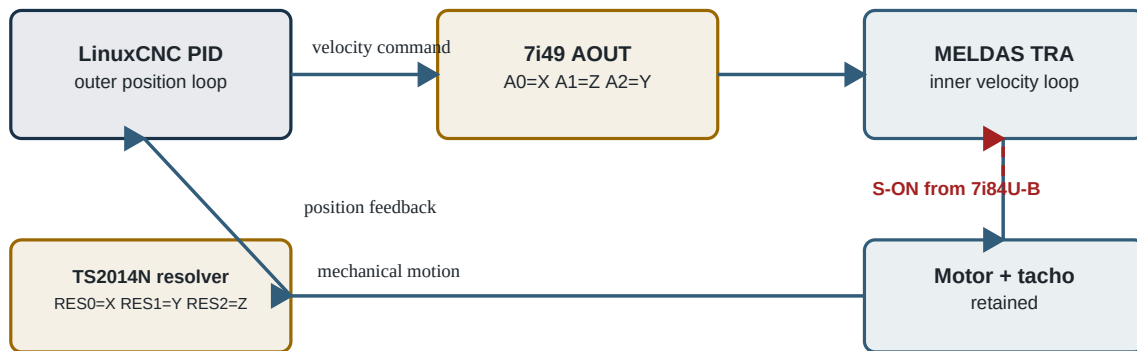


Figure 3-2. LinuxCNC closes the outer position loop; the retained TRA drive closes its inner velocity loop.

- Read and photograph the complete TS2014N suffix on X, Y, and Z. TS2014N141E26 values are comparison data only until suffixes are confirmed.
- Ohmmeter all winding pairs and compare with the exact datasheet. Do not trust old MELDAS signal names or colors.
- Use the 7i49 as the sole excitation source. The working 5 kHz selection is closest to a tentative 4.5 kHz variant but has no sourced tolerance acceptance.
- Scope RESDRV and SIN/COS at the card and resolver, at rest and through approved manual motion. Record amplitude, phase, offset, clipping, and noise.
- W2 affects only channels 3/4/5 and cannot correct X/Y/Z on channels 0/1/2.
- Derive signed resolver scale in machine units per resolver electrical revolution from screw travel and independent measurement. Do not leave 1.0 for motion.
- Calibrate volts-per-speed before tuning. OUTPUT_SCALE = 10 x measured velocity-per-volt; FF1 first, then P, with I/D only if evidence requires them.

Axis	Resolver	Analog output	Configured placeholders
X / J0	RES0	AOUT0 / pwmgen.00	res scale 1.0; output scale 1.0; gains zero
Z / J2	RES2	AOUT1 / pwmgen.01	res scale 1.0; output scale 1.0; gains zero
Y / J1	RES1	AOUT2 / pwmgen.02	res scale 1.0; output scale 1.0; gains zero
Spindle	Unassigned	AOUT3 / pwmgen.03	output scale 4000; command mode unverified

5. 7i84U field power and smart serial

Each 7i84U uses the conservative 5-28 VDC limit from the manual description/specification table. The project plans 24 VDC. Both VFIELDA and VFIELDB must be powered for their respective banks. Field power must not be mechanically switched in a way that violates the card's ramp requirement; use the documented output-enable/watchdog behavior and approved upstream design.

TB1 terminal	Function	Bank served
1 and 2	VFIELDB - both positive field supply terminals	TB2: IN16-31, OUT8-15
3 and 4	VFIELDA - both positive field supply terminals	TB3: IN0-15, OUT0-7
5	VIN logic power	W1 left links VIN to VFIELDB; W1 right needs separate VIN
6, 7, and 8	GND / VIN-VFIELD common	The only TB1 terminals used for 0 V return

TB1 POLARITY

Pins 2 and 4 are not returns. They are duplicate positive VFIELDB and VFIELDA terminals. Record each card's W1 position and never double-feed VIN.

6. Configuration files

File	Role
linuxcnc/mazak_vqc_20_40.ini	Machine/joint/spindle placeholders, ATC dry-run, timeouts, HAL load order
linuxcnc/mazak_vqc_20_40.hal	hm2_eth load, component scheduling, watchdog/E-stop/permit logic
linuxcnc/motion_7i80hdt.hal	7i49 resolver feedback, PID outer loops, pwmgen/analog outputs
linuxcnc/field_7i84u.hal	Both 7i84U pin bindings, limits, faults, enables, Z-brake sequence
linuxcnc/atc_orient.hal	ATC/orient component nets and output scheduling
linuxcnc/components/*.comp	Custom orient and ATC state machines
mesa/current_pin_authority.csv	Single physical channel authority; all wiring tables derive here

7. Staged bring-up

Stage	Required result
1	Complete applicable D1-D16 hold-point evidence; no live power before signatures.
2	Verify 7i80HDT communications, exact firmware/IDROM, HAL pins, NIC qualification, and fail-latched link-loss response.
3	Verify both 7i84U power maps, W1 positions, isolated 24 V buses, branch fusing, input modes, output load interfaces, and watchdogs.
4	With S-ON physically open, identify and energize resolver feedback only; scope all three channels and prove signed scale/direction manually.
5	With drives inhibited, verify AOUT0-AOUT3 zero voltage, polarity, mode, and measured volts-per-speed path.
6	Prove hardware E-stop, limits, drive faults, READY propagation, shared-bus stop/discharge, and Z-brake timing.
7	Execute the signed one-axis first-move plan, one axis physically enabled, at measured low authority. Do not assume Z first.
8	Tune each velocity-mode outer loop from measured scaling: FF1, P, then only evidence-driven I/D; save traces.
9	Commission spindle at low energy: analog scale, FWD/REV/RUN, zero speed, fault, gear, discrete orient, arrival/timeout cleanup.
10	Commission hydraulics and ATC output-by-output with no tool load, dry-cycle fixture, mutual exclusion, abort tests, and DRY_RUN held until acceptance.

8. Rejected and historical architectures

Item	Disposition
Mesa 7i97T	Retracted; not part of this analog-velocity/resolver stack
Mesa 7i80HD / 7i80HD-16	Different FPGA generation; never use its bitfile on a 7i80HDT
Mesa 7i37TA on P3	Retracted; duplicates field I/O and is not in the selected plan
Probe on P3 gpio.042	Retracted; unsafe 24 V on bare 3.3 V GPIO. Probe is 7i84U-B IN15
Analog orient reference on AOUT4	Retracted; ORCM1 is discrete and AOUT4 is spare
Quadrature axis encoders	Not selected; X/Y/Z feedback is resolver through 7i49